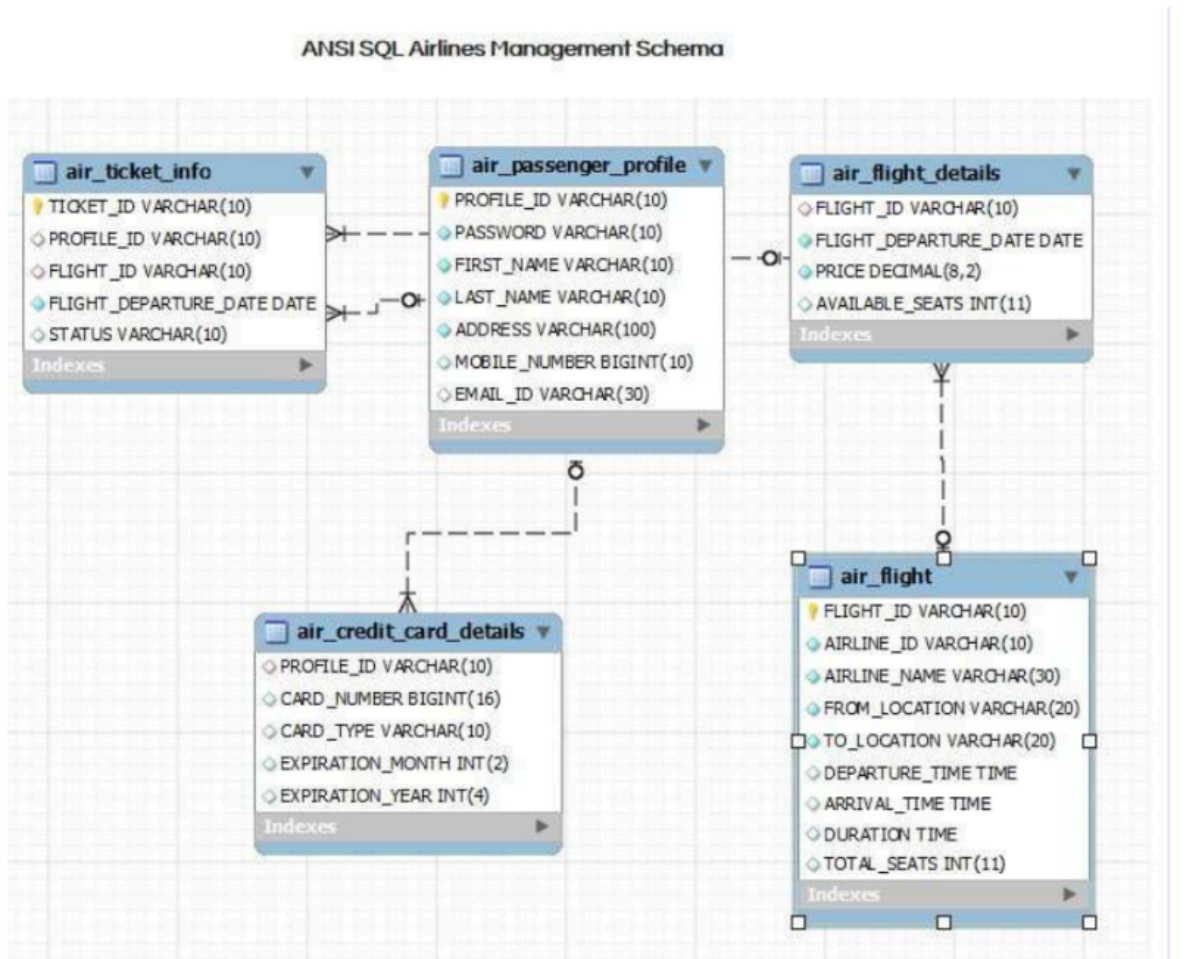


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Day 3 - Daily Exercises – SQL Server

Date: 30-12-2025 à Day 3 – Subqueries, Set Operators & CTEs



1. Write a query to display the average monthly ticket cost for each flight in ABC Airlines. The query should display the Flight_Id, From_location, To_Location, Month Name as "Month_Name" and average price as "Average_Price". Display the records sorted in ascending order based on flight id and then by Month Name.

SELECT

f.flight_id,

f.from_location,

f.to_location,

DATENAME(MONTH, s.departure_date) AS Month_Name,

AVG(s.ticket_price) AS Average_Price

FROM Flights f

JOIN Flight_Schedule s ON f.flight_id = s.flight_id

GROUP BY f.flight_id, f.from_location, f.to_location, DATENAME(MONTH, s.departure_date)

ORDER BY f.flight_id, Month_Name;

Results		Messages			
	flight_id	from_location	to_location	Month_Name	Average_Price
1	101	Chennai	Hyderabad	April	4650
2	102	Chennai	Delhi	March	6200
3	103	Mumbai	Hyderabad	April	5100
4	104	Bangalore	Chennai	February	3900

2. Write a query to display the customer(s) who has/have booked least number of tickets in ABC Airlines. The Query should display profile_id, customer's first_name, Address and Number of tickets booked as "No_of_Tickets". Display the records sorted in ascending order based on customer's first name.

```
SELECT
    c.profile_id,
    c.first_name,
    c.address,
    SUM(b.tickets) AS No_of_Tickets
FROM Customers c
JOIN Bookings b ON c.profile_id = b.profile_id
GROUP BY c.profile_id, c.first_name, c.address
HAVING SUM(b.tickets) = (
    SELECT MIN(tcount)
    FROM (
        SELECT SUM(tickets) AS tcount
        FROM Bookings
        GROUP BY profile_id
    ) x
)
ORDER BY c.first_name;
```

Results		Messages		
	profile_id	first_name	address	No_of_Tickets
1	6	Bharath	Delhi	1
2	4	Mani	Vijayawada	1
3	2	Manoj	Hyderabad	1

3. Write a query to display the number of flight services between locations in a month. The Query should display From_Location, To_Location, Month as "Month_Name" and number of flight services as "No_of_Services". Hint: The Number of Services can be calculated from the number of scheduled departure dates of a flight. The records should be displayed in ascending order based on From_Location and then by To_Location and then by month name.

```
SELECT
    f.from_location,
    f.to_location,
    DATENAME(MONTH, s.departure_date) AS Month_Name,
    COUNT(s.departure_date) AS No_of_Services
FROM Flights f
JOIN Flight_Schedule s
    ON f.flight_id = s.flight_id
GROUP BY
    f.from_location,
    f.to_location,
    DATENAME(MONTH, s.departure_date)
ORDER BY
    f.from_location,
    f.to_location,
    Month_Name;
```

	from_location	to_location	Month_Name	No_of_Services
1	Bangalore	Chennai	February	1
2	Chennai	Delhi	March	1
3	Chennai	Hyderabad	April	2
4	Mumbai	Hyderabad	April	1

4. Write a query to display the customer(s) who has/have booked maximum number of tickets in ABC Airlines. The Query should display profile_id, customer's first_name, Address and Number of tickets booked as "No_of_Tickets". Display the records in ascending order based on customer's first name.

```
SELECT
    c.profile_id,
    c.first_name,
    c.address,
    SUM(b.tickets) AS No_of_Tickets
FROM Customers c
JOIN Bookings b
    ON c.profile_id = b.profile_id
```

GROUP BY

c.profile_id,

c.first_name,

c.address

HAVING SUM(b.tickets) = (

SELECT MAX(total_tickets)

FROM (

SELECT SUM(tickets) AS total_tickets

FROM Bookings

GROUP BY profile_id

) x

)

ORDER BY c.first_name;

	profile_id	first_name	address	No_of_Tickets
1	1	Sanjay	Chennai	3
2	3	Varshith	Bangalore	3

5. Write a query to display the number of tickets booked from Chennai to Hyderabad. The Query should display passenger profile_id,first_name,last_name, Flight_Id , Departure_Date and number of tickets booked as "No_of_Tickets". Display the records sorted in ascending order based on profile id and then by flight id and then by departure date.

SELECT

c.profile_id,

c.first_name,

c.last_name,

b.flight_id,

b.departure_date,

b.tickets AS No_of_Tickets

FROM Bookings b

JOIN Customers c

ON b.profile_id = c.profile_id

JOIN Flights f

ON b.flight_id = f.flight_id

WHERE f.from_location = 'Chennai'

AND f.to_location = 'Hyderabad'

```
ORDER BY
  c.profile_id,
  b.flight_id,
  b.departure_date;
```

	profile_id	first_name	last_name	flight_id	departure_date	No_of_Tickets
1	1	Sanjay	Kumar	101	2025-04-10	2
2	1	Sanjay	Kumar	101	2025-04-21	1
3	4	Mani	Kumar	101	2025-04-10	1

6. Write a query to display flight id,from location, to location and ticket price of flights whose departure is in the month of april.

```
SELECT
  f.flight_id,
  f.from_location,
  f.to_location,
  s.ticket_price
FROM Flights f
JOIN Flight_Schedule s
  ON f.flight_id = s.flight_id
WHERE MONTH(s.departure_date) = 4;
```

	flight_id	from_location	to_location	ticket_price
1	101	Chennai	Hyderabad	4500
2	101	Chennai	Hyderabad	4800
3	103	Mumbai	Hyderabad	5100

7. Write a query to display the average cost of the tickets in each flight on all scheduled dates. The query should display flight_id, from_location, to_location and Average price as "Price". Display the records sorted in ascending order based on flight id and then by from_location and then by to_location.

```
SELECT

  f.flight_id,

  f.from_location,

  f.to_location,

  AVG(s.ticket_price) AS Price

FROM Flights f

JOIN Flight_Schedule s

  ON f.flight_id = s.flight_id
```

GROUP BY

```
f.flight_id,  
f.from_location,  
f.to_location
```

ORDER BY

```
f.flight_id,  
f.from_location,  
f.to_location;
```

Results

Messages

	flight_id	from_location	to_location	Price
1	101	Chennai	Hyderabad	4650
2	102	Chennai	Delhi	6200
3	103	Mumbai	Hyderabad	5100
4	104	Bangalore	Chennai	3900

8. Write a query to display the customers who have booked tickets from Chennai to Hyderabad. The query should display profile_id, customer_name (combine first_name & last_name with comma in b/w), address of the customer. Give an alias to the name as customer_name. Hint: Query should fetch unique customers irrespective of multiple tickets booked. Display the records sorted in ascending order based on profile id.

SELECT DISTINCT

```
c.profile_id,  
c.first_name + ',' + c.last_name AS customer_name,  
c.address
```

FROM Customers c

JOIN Bookings b

```
ON c.profile_id = b.profile_id
```

JOIN Flights f

```
ON b.flight_id = f.flight_id
```

WHERE f.from_location = 'Chennai'

```
AND f.to_location = 'Hyderabad'
```

ORDER BY c.profile_id;

Results		Messages	
	profile_id	customer_name	address
1	1	Sanjay,Kumar	Chennai
2	4	Mani,Kumar	Vijayawada

9. Write a query to display profile id of the passenger(s) who has/have booked maximum number of tickets. In case of multiple records, display the records sorted in ascending order based on profile id.

```

SELECT profile_id
FROM Bookings
GROUP BY profile_id
HAVING SUM(tickets) = (
    SELECT MAX(total_tickets)
    FROM (
        SELECT SUM(tickets) AS total_tickets
        FROM Bookings
        GROUP BY profile_id
    ) y
)
ORDER BY profile_id;

```

Results		Messages	
	profile_id		
1	1		
2	3		

10. Write a query to display the total number of tickets as "No_of_Tickets" booked in each flight in ABC Airlines. The Query should display the flight_id, from_location, to_location and the number of tickets. Display only the flights in which atleast 1 ticket is booked. Display the records sorted in ascending order based on flight id.

```

SELECT
    f.flight_id,
    f.from_location,

```

f.to_location,

SUM(b.tickets) AS No_of_Tickets

FROM Flights f

JOIN Bookings b

ON f.flight_id = b.flight_id

GROUP BY

f.flight_id,

f.from_location,

f.to_location

HAVING SUM(b.tickets) >= 1

ORDER BY f.flight_id;

Results		Messages		
	flight_id	from_location	to_location	No_of_Tickets
1	101	Chennai	Hyderabad	4
2	102	Chennai	Delhi	1
3	103	Mumbai	Hyderabad	4
4	104	Bangalore	Chennai	2